The Topology of the Path Space

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Abstract

These notes are prepared for my second talk in the student seminar *Introduction to Morse Theory*. We prove some results about the topology of the path space of a Riemannian manifold following the book [4] by Milnor.

1. Introduction

1.1. *The setup.* Let (M, g) be a Riemannian manifold with the induced topological metric ρ. Let p and q be two not necessarily distinct points of M. There are two spaces of paths on M which we are interested in. The first one is the space $\Omega^* = \Omega^*(M; p, q)$ of all continuous paths $\omega : [0, 1] \to M$ with $\omega(0) = p$ and $\omega(1) = q$, equipped with the metric

$$d^*(\omega_1, \omega_2) = \max_{t\in[0,1]} \rho(\omega_1(t), \omega_2(t)) \qquad \forall \omega_1, \omega_2 \in \Omega^*.$$

The second one is the space $\Omega = \Omega(M; p, q)$ of piecewise smooth paths from p to q with the metric

$$d(\omega_1, \omega_2) = d^*(\omega_1, \omega_2) + \left(\int_0^1 \left((s_1'(t) - s_2'(t) \right)^2 \, dt \right)^{1/2} \qquad \forall \omega_1, \omega_2 \in \Omega,$$

where for $\omega \in \Omega$,

$$s(t) = \int_0^t \|\dot{\omega}(\tau)\| \, d\tau.$$

Since $d \geq d^*$, the natural inclusion $\iota : \Omega \hookrightarrow \Omega^*$ is continuous. It turns out that we can further exploit nice properties of the inclusion, which will be the topic of Section 2. In Section 3 we will see that under mild assumptions Ω is homotopy equivalent to a countable CW-complex. Finally, we will conclude with a few results about the path space of the sphere as an example in Section 4.

2. Homotopy equivalence between Ω and Ω^*

2.1. *Main result.* The main theorem we will prove in this section is

Theorem 2.1. *The inclusion $\iota : \Omega \hookrightarrow \Omega^*$ is a homotopy equivalence.*

2.2. *Preliminary results.* In the proof we will need some results which we do not prove. We start with [2, Theorem 6.17]:

THEOREM 2.2. *Let (M, g) be a Riemannian manifold. For each $p \in M$, there exists $\varepsilon > 0$ such that every geodesic ball centred at p of radius less than or equal to ε is geodesically convex.*

We will be dealing with the homotopy type of a countable union of topological spaces in the proof. More specifically, let X be a topological space and $X_1 \subseteq X_2 \subseteq X_3 \subseteq \cdots$ be a sequence of subspaces. Let

$$X_\Sigma = \bigcup_{i=1}^{\infty} X_i \times [i, i+1] \subseteq X \times \mathbb{R}.$$

We introduce the following notion:

DEFINITION 2.3. X is the *homotopy direct limit* of the sequence (X_i) if the projection on to the first coordinate $\pi : X_\Sigma \to X$ is a homotopy equivalence.

The next theorem says that the homotopy type of X is almost completely determined by the homotopy type of the X_i. See [4, Theorem A] for a proof.

THEOREM 2.4. *Suppose that X is the homotopy direct limit of (X_i) and Y is the homotopy direct limit of (Y_i). Let $f : X \to Y$ be a map which carries each X_i into Y_i by a homotopy equivalence. Then f itself is a homotopy equivalence.*

2.3. *Proof of Theorem 2.1.* Now we are able to prove the main theorem using the tools we introduced.

Proof of Theorem 2.1. We cover M by geodesically convex open sets $(N_\alpha)_{\alpha \in A}$ given by Theorem 2.2. Divide $[0, 1]$ into 2^k subintervals $[(j-1)2^{-k}, j2^{-k}]$ and let Ω_k^* denote the set of all continuous paths ω from p to q which satisfy $\omega([(j-1)2^{-k}, j2^{-k}]) \subseteq N_\alpha$ for some $\alpha \in A$ for all $1 \leq k$. Clearly Ω_k^* is an open subset of Ω^* because N_α is open and $\omega([(j-1)2^{-k}, j2^{-k}])$ is compact. With this construction we have $\Omega_1^* \subseteq \Omega_2^* \subseteq \cdots \subseteq \Omega_{k-1}^* \subseteq \Omega_k^* \subseteq \cdots$ and

$$\Omega^* = \bigcup_{k=1}^{\infty} \Omega_k^*.$$

Similarly, $\Omega_k := \iota^{-1}(\Omega_k^*)$ are ope sets of Ω and

$$\Omega = \bigcup_{k=1}^{\infty} \Omega_k.$$

The next goal is to show that $\iota|_{\Omega_k} : \Omega_k \to \Omega_k^*$ is a homotopy equivalence. Define $h : \Omega_k^* \to \Omega_k$ as follows: For each $\omega \in \Omega_k^*$, let $h(\omega)$ be the broken geodesic with $h(\omega)(t) = \omega(t)$ for $t = j2^{-k}$, $0 \leq j \leq 2^k$ and such that $h(\omega)|_{[(j-1)2^{-k}, j2^{-k}]}$ is a

minimal geodesic connecting the endpoints. h is continuous because a geodesic depends smoothly on its endpoints (see [1, Remark 3.8]). $\iota|_{\Omega_k} \circ h$ maps ω to $h(\omega)$, and since both $\omega([(j-1)2^{-k}, j2^{-k}])$ and $h(\omega)([(j-1)2^{-k}, j2^{-k}])$ lie in a geodesically convex set N_α they are homotopic to each other via a linear homotopy in $\mathbb{R}^n$ and a local chart[1]. Hence $\iota|_{\Omega_k} \circ h \simeq \mathrm{id}_{\Omega_k^*}$. $h \circ \iota|_{\Omega_k}$ replaces each segment $\omega([(j-1)2^{-k}, j2^{-k}])$ by a geodesic segment while fixing the endpoints, so using the same argument we obtain $h \circ \iota|_{\Omega_k} \simeq \mathrm{id}_{\Omega_k}$.

In order to conclude the proof, it remains to show that Ω is a homotopy direct limit of Ω_k, and the same for Ω^*. Let $\pi : \Omega_\Sigma \to \Omega$ be the projection as in Definition 2.3. Since Ω is a metric space, by [5, Theorem 41.4] it is paracompact[2]. A metric space is also Hausdorff, so [5, Theorem 41.7] gives a partition of unity $\{\lambda_k\}$ subordinated to the open cover $\{\Omega_k\}$. Define $f : \Omega \to \mathbb{R}$ by

$$f = \sum_{k=1}^{\infty} k\lambda_k.$$

Then $f \geq 0$ and

$$f(\omega) = \sum_{k=j+1}^{\infty} k\lambda_k \geq j+1$$

if $\omega \notin \Omega_j$. Thanks to this function, we can define $g : \Omega \to \Omega_\Sigma$ by $\omega \mapsto (\omega, f(\omega))$, g is clearly continuous and satisfies $\pi \circ g = \mathrm{id}_\Omega$. We claim that $g \circ \pi$ is a deformation retraction. Indeed, an explicit homotopy between $\mathrm{id}_{\Omega_\Sigma}$ and $g \circ \pi$ is given by

$$\begin{aligned} H : [0,1] \times \Omega_\Sigma &\longrightarrow \Omega_\Sigma \\ (s, (\omega, x)) &\longmapsto (\omega, (1-t)x + tf(x)). \end{aligned}$$

This proves that π is a homotopy equivalence, hence Ω is a homotopy direct limit of Ω_k. The exact same argument applies to the metric space Ω^*.

Having shown that each $\iota|_{\Omega_k}$ is a homotopy equivalence, we can invoke Theorem 2.4 to conclude that ι is also a homotopy equivalence. □

[1] Up to assuming that each N_α is simply connected.

[2]

Definition 2.5. A topological space X is *paracompact* if every open covering $\mathcal{A}$ of X has a locally finite open refinement $\mathcal{B}$ that covers X.

3. **Fundamental theorem of Morse Theory**

In this section, we will give a more precise description of the topology of the path space Ω. To begin with, let us review a result proven by Milnor in [3]. Let $\mathcal{W}$ denote all topological spaces with the homotopy type of a CW-complex. Then, we have

THEOREM 3.1. *If A belongs to the class $\mathcal{W}$ and C is compact, then the function space of continuous maps A^C also belongs to $\mathcal{W}$.*

Combining Theorems 2.1 and 3.1 gives

COROLLARY 3.2. *Ω has the homotopy type of a CW-complex.*

A more precise statement is:

THEOREM 3.3 (Fundamental theorem of Morse Theory). *Let M be a* complete *Riemannian manifold, and let $p, q \in M$ be two points which are not conjugate along any geodesic. Then $\Omega(M; p, q)$ and $\Omega^*(M; p, q)$ have the homotopy type of a countable CW-complex which contains a cell of dimension λ for each geodesic from p to q whose index form has index λ.*

Proof sketch. Choose a sequence $a_0 < a_1 < a_2 < \cdots$ which are not critical points of the energy function E, so that each interval (a_{i-1}, a_i) contains precisely one critical value[3]. Consider the sequence

$$\Omega_0 \subseteq \Omega_1 \subseteq \Omega_2 \subseteq \cdots ,$$

where $\Omega_i = E^{-1}((-\infty, a_i))$. Using Theorem 16.1, Remark 3.3 and Lemma 3.7 from [4] we obtain that each Ω_i is homotopy equivalent to Ω_{i-1} with a finite number of cells attached: one λ-cell for each geodesic with index λ. As in the proof of [4, Theorem 3.5] we construct homotopy equivalences $f_i : \Omega_i \to K_i$ compatible with with the inclusions, where K_i are CW-complexes and obtain the following commutative diagram:

$$\begin{tikzcd}
\Omega_0 \arrow[r, hook] \arrow[d, "f_0"] & \Omega_1 \arrow[r, hook] \arrow[d, "f_1"] & \Omega_2 \arrow[r, hook] \arrow[d, "f_2"] & \cdots \\
K_0 \arrow[r, hook] & K_1 \arrow[r, hook] & K_2 \arrow[r, hook] & \cdots
\end{tikzcd}$$

Let K be the direct limit (colimit) $K = \varinjlim K_i$, similarly $\Omega = \varinjlim \Omega_i$. Denote by $f : \Omega \to K$ the map induced from f_i. Now we can apply Theorem 2.4 to conclude because

[3]This is not trivial unless we know the set of critical values is discrete. To see this, consider the finite dimensional approximation from [4, Theorem 16.2]: Since p and q are not conjugate along any geodesic, the critical points of E, and thus of E', are non-degenerate (any admissible variation vector field for E' is also admissible for E). We may now apply the results for finite dimensions to E' to conclude

(1) Ω is a homotopy direct limit using the same argument from the proof of Theorem 2.1.
(2) For X we claim that the projection $\pi : X_\Sigma \to X$ induces isomorphisms of homotopy groups in all dimensions. First note that the image of any $[\sigma] \in \pi_n(X_\Sigma)$ is compact, so it is fully contained in a compact subset $X_1 \times [1,2] \cup \cdots \cup X_N \times [N, N+1]$. Using homotopies in the second coordinate, we can move σ to $X_N \times \{N+1\}$ to obtain $\tau \simeq \sigma$ with $[\tau] \in \pi_n(X_N \times [N, N+1]) \cong \pi_n(X_N) \leq \pi_n(X)$. Hence $\pi_* : \pi_n(X_\Sigma) \to \pi_n(X)$ is surjective. If $\pi_*([\sigma]) = \pi_*([\tau])$, we move both to $\sigma', \tau' \in X_N \times \{N+1\}$. We retain the equality $\pi_*([\sigma']) = \pi_*([\tau'])$, but now the homotopy between $\pi_*([\sigma'])$ and $\pi_*([\tau'])$ gives directly a homotopy between σ' and τ', which shows injectivity. Therefore, π_* is an isomorphism for all π_n, now by Whitehead's [6, Theorem 1], π is a homotopy equivalence and X is a homotopy direct limit.

This ends our proof. □

4. **The path space of S^n**

Let $p, q \in S^n$ be two non-conjugate points, equivalently, $q \neq p, p'$, where p' is the antipode of p. We will see why this equivalence holds from an argument we use for a later proof. There are countably many geodesics $\gamma_0, \gamma_1, \gamma_2, \ldots$ from p to q: γ_0 is the short great circle arc from p to q; γ_1 is the long great circle arc $pq'p'q$; γ_2 is the arc $pqp'q'pq$; etc. The subscript in γ_k indicate the total number of times p and p' occur in the *interior* of γ_k.

Proposition 4.1. *Let $\lambda(\gamma_k)$ be the index of γ_k. Then $\lambda(\gamma_k) = k(n-1)$.*

Proof. Thanks to the Morse Index Theorem, it suffices to count the number of conjugate points with multiplicity. We claim that the only points conjugate to p are p and p'. Suppose another point p'' is conjugate to p along either the short or long great circle arc γ (these are the only geodesics), then by [4, Lemma 14.4], any Jacobi field along γ can be obtained from a variation of it through geodesics. In particular, there exists geodesics arbitrarily close to γ, which is obviously false. Next, we count the index of p and p'. Using [4, Lemmas 14.3 and 14.4], this is equivalent to counting the variations of γ through geodesics which produce linearly independent Jacobi fields.

For p', any geodesic c from p to p' is uniquely determined by the intersection of c with the equator (when p is taken to be the north pole), hence the set of geodesics near γ (equipped with the L^∞ norm) is homeomorphic to D^{n-1}. Thus, each axis of D^{n-1} corresponds to a variation of γ through geodesics. These variations give rise to Jacobi fields which are orthogonal to each other

at the intersection of γ with the equator, hence linearly independent. [4, Remark 14.2] gives the upper bound $n-1$ for the dimension of the space of Jacobi fields, hence the multiplicity is exactly $n-1$.

The same argument works for p as a conjugate point of p. From here, the proposition follows directly. □

An immediate consequence is

Corollary 4.2. *The path space $\Omega(S^n)$ has the homotopy type of a CW-complex with one cell each in the dimensions $0, n-1, 2(n-1), 3(n-1), \ldots$*

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